

Fisher's inequality excludes two proposed Steiner ETF candidates

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Status. Literature-implied answer, complete for the two finite candidates. The broader ETF restricted-isometry question remains open.

1 Source question

Fickus, Mixon, and Tremain [1], at the end of Section 3 on page 14, write that they were unable to resolve two remaining candidates: the 23×276 and 46×736 real equiangular tight frames could potentially arise from $(2, 10, 46)$ - and $(2, 14, 92)$ -Steiner systems, respectively, provided those systems exist.

2 Identification

The classical Fisher inequality for block designs settles both cases. Horsley [2, p. 2] recalls that every nontrivial 2 -(v, k, λ) design has at least v blocks, equivalently

$$\lambda(v-1) \geq k(k-1).$$

Proposition 1. *Neither a $(2, 10, 46)$ -Steiner system nor a $(2, 14, 92)$ -Steiner system exists. Consequently, neither of the two source ETFs can arise from the proposed Steiner construction.*

Proof. Let A be the $v \times b$ incidence matrix of a nontrivial 2 -(v, k, λ) design, and let r be its replication number. Then

$$AA^T = (r - \lambda)I + \lambda J.$$

Because $v > k$, one has $r > \lambda$. The displayed matrix is therefore positive definite: its eigenvalues are $r - \lambda$, with multiplicity $v - 1$, and $r - \lambda + \lambda v$, with multiplicity one. Hence

$$v = \text{rank}(AA^T) \leq \text{rank}(A) \leq b,$$

which is Fisher's inequality.

For a $(2, k, v)$ -Steiner system, $\lambda = 1$ and double counting gives

$$b = \frac{v(v-1)}{k(k-1)}.$$

The first proposed system would have

$$b = \frac{46 \cdot 45}{10 \cdot 9} = 23 < 46,$$

and the second would have

$$b = \frac{92 \cdot 91}{14 \cdot 13} = 46 < 92.$$

Both contradict $b \geq v$. □

3 Scope and provenance

The argument gives a complete negative answer to the two explicit finite candidates and also records the general necessary condition

$$v - 1 \geq k(k - 1)$$

for every nontrivial $(2, k, v)$ -Steiner system. This is not a new theorem: it is a direct application of Fisher’s classical inequality, apparently missed in the source discussion. It is therefore classified as a literature-implied answer rather than as a new full solution.

The source’s final question—whether some other equiangular tight frames can have restricted-isometry behavior beyond the square-root bottleneck—is not answered here.

References

- [1] M. Fickus, D. G. Mixon, and J. C. Tremain, *Steiner equiangular tight frames*, Linear Algebra Appl. **436** (2012), 1014–1027; [arXiv:1009.5730](#).
- [2] D. Horsley, *Generalising Fisher’s inequality to coverings and packings*, Australas. J. Combin. **70** (2018), 339–372; [arXiv:1409.0485](#).